

Temporal trends of persistent organic pollutants in Barents Sea polar bears (*Ursus maritimus*) in relation to changes in feeding habits and body condition

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CONTEXT :

- Dominant **pollutant** in Arctic :
 - persistent organic pollutants (**POPs**)
 - transported over **long distances** via atmospheric, oceanic and river currents
- **Bioaccumulation** :
 - high concentrations in **top predators**
 - **health effects** in animals and humans :
hormonal, immune, metabolic disruptions

OBJECTIVES :



- ◆ Study temporal trend in POP in Barents Sea polar Bears (1997-2017)
- ◆ Explore the links between [POP], dietary and body condition
- ◆ Link those changes with Climate change

MATERIALS AND METHODS :

Sampling over **20 years**, 306 samples and **185 individuals**

- Feeding habits : **stable isotopes** of carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$)
 - distinguish **marine** and **terrestrial prey** + estimate **trophic level**
- Body Condition Index (BCI) calculated from :
 - **estimated body mass + length**
- POPs: **15 major compounds** analyzed
 - PCBs, OH-PCBs, OCP and PBDEs

BCI
Feeding habits
POPs

**Generalized additive
mixed model
(GAMM)**

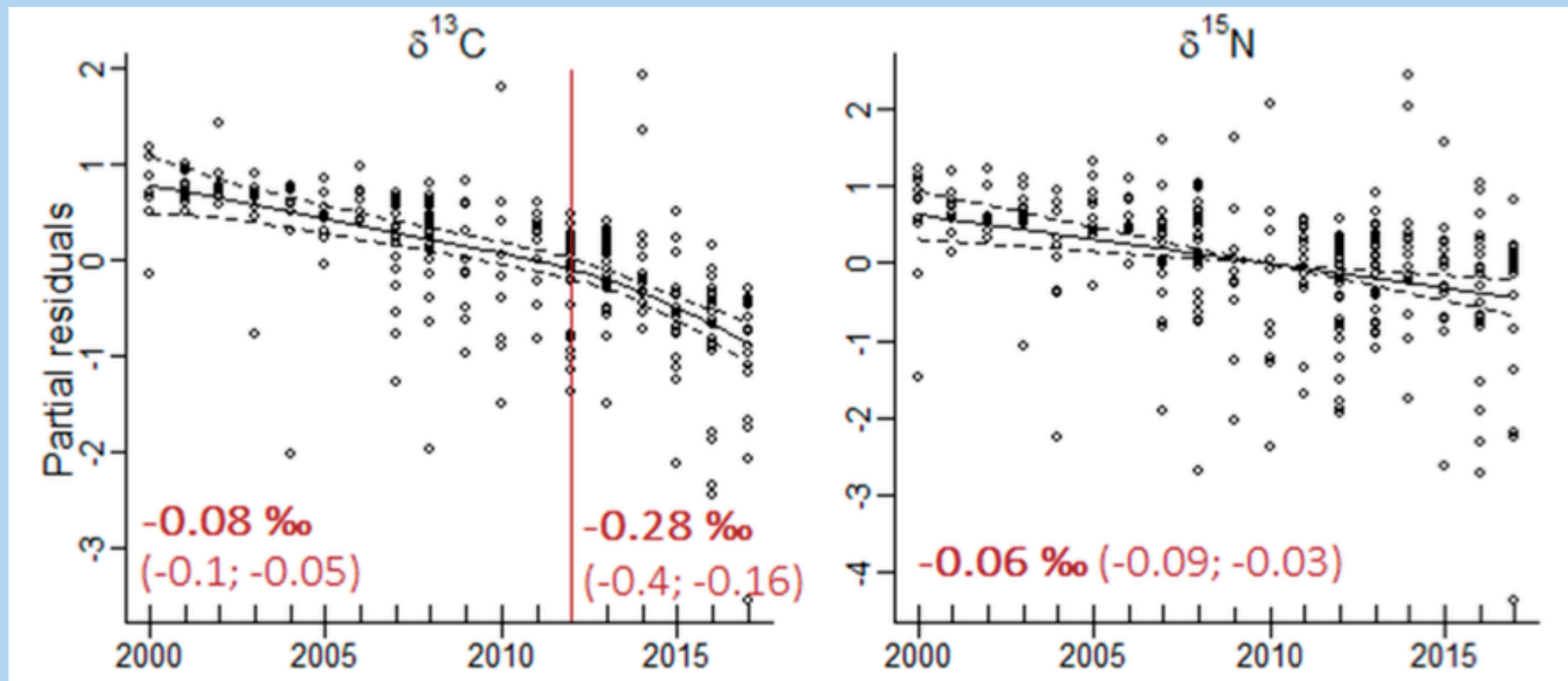
**Link + temporal
changes of feeding
habits, BCI and [POPs]**

RESULTS AND CONCLUSION :



- ◆ Feeding habits : **reduction** in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values over 20 years
 - **less marine** and more terrestrial diet
 - feeding at **lower trophic levels**

*probably linked
to **melting sea
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- ◆ [POPs] : Significant **decline**:

- * PCBs (-5%/year)
- * OH-PCBs (-6%/year)

*probably due to **regulatory efforts**
(Stockholm Convention)*

Recent **increases** :

- * OCPs (+8 to 21%/year after 2009)

*possibly due to **secondary
emissions** (thawing, volatilization)
or **residual** primary emissions*

Limited influence of dietary and trophic changes on POPs concentrations

- ◆ BCI : **Decline** between 1997 and 2005
Significant **increase** since 2005

*suggesting an **adaptation**
to new dietary conditions*

DISCUSSION AND PERSPECTIVES :

Regulatory success:

POPs reduced through **international efforts**
BUT **climate change** raises new concerns

Impact of global warming:

Thawing ice and permafrost → Potential **re-emission** of POPs
Habitat and diet disruption → Potential effects on future [POPs]

Critical monitoring:

Continue **monitoring** of POPs, dietary changes, and body condition of bears
Anticipate impacts on polar bear **reproduction** and **survival**

Thank you for listening !

Some questions ?



Critical Analysis of the Study :

- Only **adult females** were sampled
—> excluding **sex-related** variation
- The use of **blood plasma** and not adipose tissue
—> reflects recent exposures but does not allow assessment of long-term accumulation
- **Complex interactions** between diet, body conditions, and environmental factors
—> could affect the results
- **Short-term** or **seasonal trends** may not be captured with annual sampling.